Responses: Training Data, Hypothesis, and Neural Network Model for Predicting Comfort Temperature

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Q1: How did you make the training data?

The training dataset was generated using appropriate statistical distributions, followed by neural network training using TensorFlow Keras. The document also provides details on the neural network architecture, optimization methods, and visual representations of data distributions.

How many training data?

To generate the training dataset, we used 1,000,000 (1 million) data points with the following distributions for the various parameters:

- **Air Velocity**: Sampled from a log-normal distribution with a scale parameter of 0.15, resulting in a realistic range of air velocities between 0.0 m/s to 0.8 m/s. This distribution was chosen because air velocity in practical settings tends to have a positive skew.
- **Air Temperature**: Sampled from a normal distribution with a mean (average) of 24°C and a standard deviation (scale) of 2°C, clipped to values between 18°C to 30°C to ensure realistic indoor temperatures.
- **Relative Humidity**: Fixed at 50% for all samples to simplify the calculations. Initially, beta-distributed values were tested, but results were similar, leading to the simplification, as well as ACs are designed to maintain the room humidity around 50%.
- **Mean Radiant Temperature (MRT)**: Set equal to the Air Temperature, which is often appropriate for indoor conditions where MRT is similar to the air temperature.
- **Metabolic Rate and Clothing Insulation**: Fixed values of 1.0 met for metabolic rate and 0.5 clo for clothing insulation were used to represent moderate physical activity and typical indoor clothing conditions.

The ASHRAE Standard 55 was used for calculating the Predicted Mean Vote (PMV) and Percentage of People Dissatisfied (PPD) for each data point, using the pythermalcomfort library. The official reference for the ASHRAE-55 PMV calculation formula is:
ASHRAE 55-2020, Thermal Environmental Conditions for Human Occupancy. Available at:

`https://www.ashrae.org/technical-resources/bookstore/standard-55-thermal-environmental-conditions-`

Data Distribution and Visualization

Below are the distributions for different air velocity and air temperature (inside room) used for generating the training dataset:

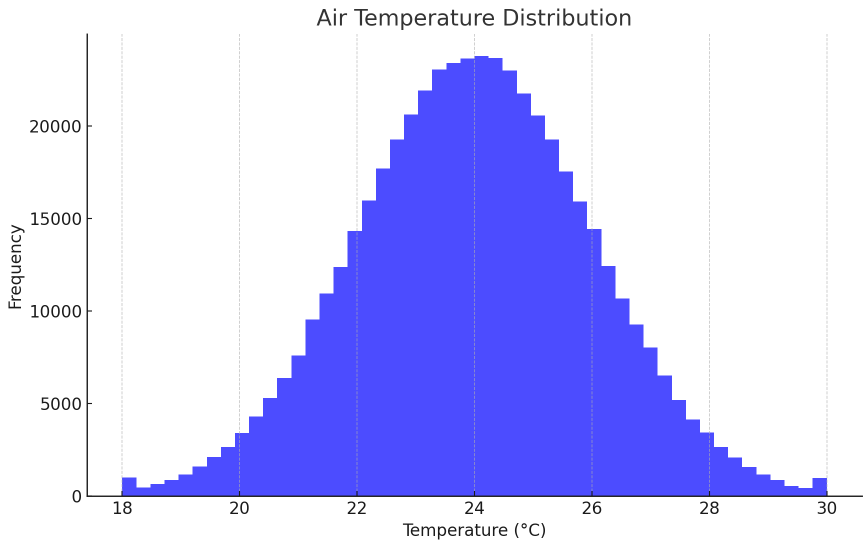


Figure 1: Air Temperature Distribution: Mean of 24°C, Std Dev of 2°C, clipped between 18°C and 30°C.

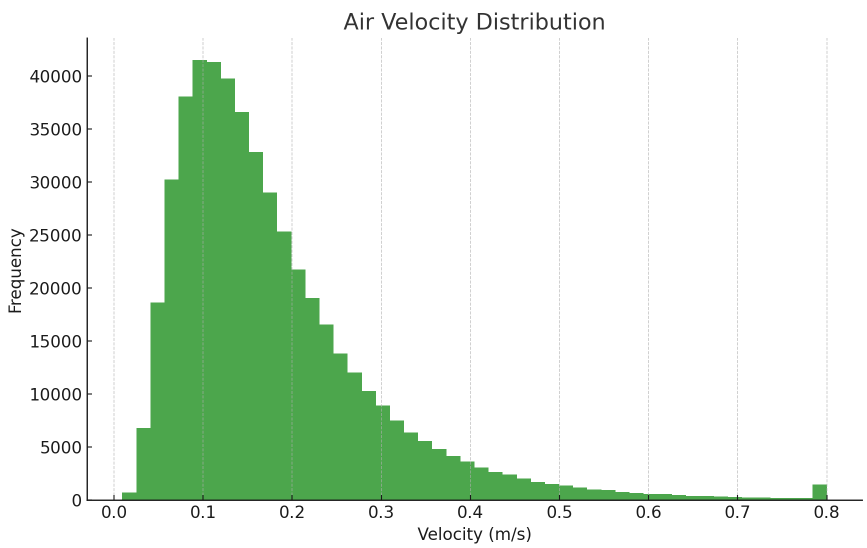


Figure 2: Air Velocity Distribution: Log-normal distribution with most samples falling below 0.8 m/s.

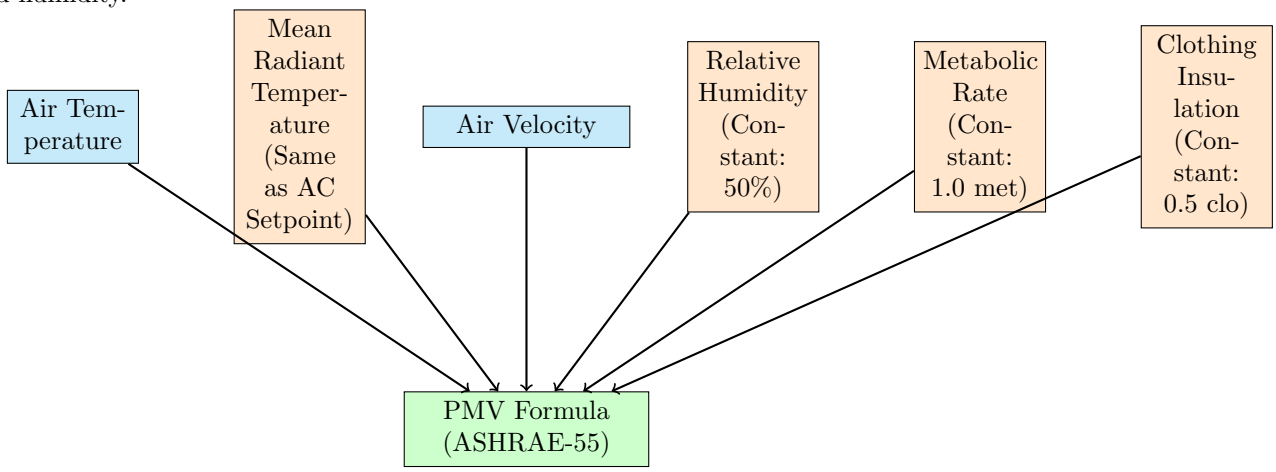
PMV Calculations

The Predicted Mean Vote (PMV) values were computed for each of the generated data points as a part of the initial data generation. For the neural network model during training, while the PMV and outdoor were used as features, air velocity and air temperature were used as the target prediction variables. The PMV function incorporates multiple non-linear relationships. The fundamental equation for PMV includes exponential terms and complex heat transfer calculations:

$$PMV = (0.303e^{-0.036M} + 0.028)L \tag{1}$$

where M is metabolic rate and L is thermal load of room, determined by the air temperature of the room

and humidity.



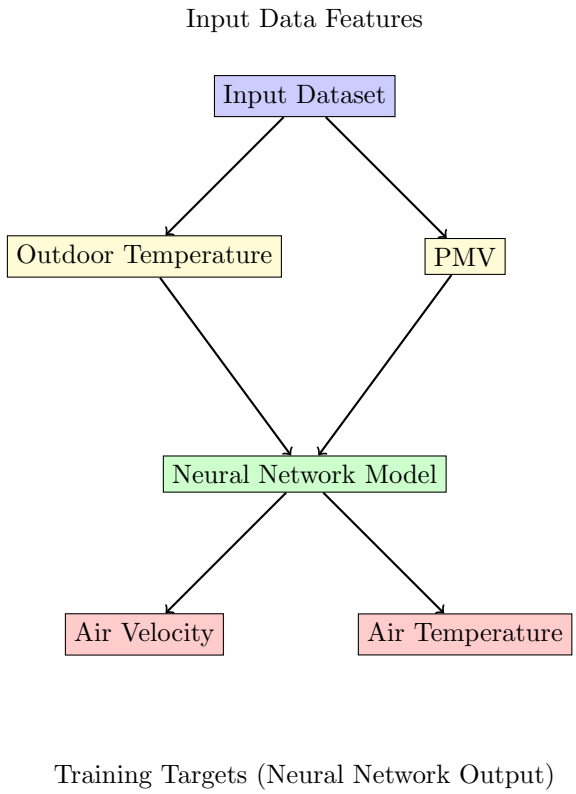
Neural Network Model

The thermal comfort prediction system operates in a complex mathematical space defined by the ASHRAE Standard 55 comfort model. At its core, the system must solve a multiobjective optimization problem where the primary variables are air temperature (T) and air velocity (v), while considering outdoor temperature (T_{out}) and the Predicted Mean Vote (PMV) index. The neural network model was trained on the generated data set to predict the comfort temperature of an AC and a fan that work in sync.

Why so few features?

Our proposed model is actually using the minimum necessary features while capturing maximum information (1) PMV and (2) Outdoor Temperature for its output predictions (1) Air Temperature and (2) Air Velocity. We believe this is efficient because PMV already encapsulates multiple comfort parameters, and considers what a AC is likely to maintain constant, and people are likely wear in the comfort of their homes. The outdoor temperature provides crucial context for system operation. Some notes:

- More features don't necessarily mean better predictions (Occam's Razor principle). The model achieves its goal with minimal complexity, which is generally preferred in machine learning
- Having just two fixed parameters reduce noise and focus the model on the most important variables for AC and fan control



Data Preparation

The dataset was filtered to keep only samples where $-0.1 \leq \text{PMV} \leq 0.1$, ensuring the data focused on comfortable thermal conditions. In addition to the columns of the training dataset, additionally a column with appropriate outdoor temperatures were added to the training data. The data was then split into 80% training and 20% testing sets. The features were scaled using StandardScaler from the 'scikit-learn' library.

Model Architecture

The neural network was implemented using TensorFlow Keras with the following architecture:

- **Input Layer**: Takes two features ("PMV" and "Air Velocity").
- **Hidden Layers**:
  - Dense Layer 1: 64 neurons with ReLU activation function.
  - Dense Layer 2: 32 neurons with ReLU activation function.
  - Dense Layer 3: 16 neurons with ReLU activation function.
- textbfOutput Layer: 1 neuron without an activation function, predicting the Air Temperature.

The model was compiled using the Adam optimizer with a learning rate of 0.001 and mean squared error (MSE) as the loss function. The Adam optimizer was selected for its adaptive learning rate capabilities, which typically lead to faster convergence and fewer manual adjustments.

The model was trained for 100 epochs with a batch size of 32. Validation split was set to 20% of the training data.

How is the model used in the service?

In the testing and real-world application phase, given a PMV value and the outside temperature, the model predicts the required Air Temperature and Air Velocity for maintaining comfortable indoor conditions.

Given the user input in the UX(app) on comfort temperature, the model adjusts its output using the neural network to predict the optimal AC Setpoint (T_{AC}) and Air Velocity (V_{room}). Tthe feature vector is $X = [PMV, T_{outdoor}, V_{room}]$, where $T_{outdoor}$ is the outdoor temperature. The neural network aims to learn a mapping function $f(X;\theta)$ to predict the indoor temperature:

$$T_{AC} = f(X;\theta), \tag{2}$$

where θ represents the parameters learned during training. By minimizing the Mean Squared Error (MSE) between predicted and desired comfort settings, the model effectively tunes itself to different AC setpoints and fan-induced air velocities during the training phase.

Q2: What is the hypothesis of your model?

The hypothesis of the model is that the thermal comfort experienced by individuals in an indoor environment can be effectively predicted based on a combination of environmental variables, specifically air velocity, air temperature, and the Predicted Mean Vote (PMV). By accurately predicting the Air Temperature and Air Velocity that maintains a desired PMV, the model aims to ensure occupant comfort while minimizing energy consumption.

The underlying assumption is that the PMV, derived from a combination of physiological and environmental factors, serves as a reliable indicator of thermal comfort. Given a fixed PMV value (which represents comfortable conditions for typical healthy adults), the neural network model should be capable of predicting the ideal Air Temperature and Air Velocity settings for an AC and fan combination, ensuring that indoor conditions remain comfortable without excessive energy use.

During the training phase, the PMV value was used as the target variable. In the testing and real-world application phase, given a PMV value and the outside temperature, the model predicts the required Air Temperature and Air Velocity for maintaining comfortable indoor conditions.

Q3: How can the model determine user preferred temperatures and make predictions?

To determine user preferred temperatures and make predictions, the model incorporates a feedback-driven approach that learns from user behavior and preferences over time. The goal is to continuously adjust the comfort settings in response to user input, ensuring that individualized comfort is achieved. This process is grounded in mathematical modeling, machine learning, and user experience (UX) design principles.

User Feedback Mechanism

The model integrates a user feedback mechanism within the User Experience (UX) interface. This feedback loop allows users to adjust their comfort temperature in real time based on their subjective thermal experience. If a user finds the temperature too hot or too cold, they can provide feedback via the application, which then fine-tunes the Air Temperature and Air Velocity settings. Once the comfort temperature is adjusted, the AI model recalculates the necessary AC setpoint and air velocity in the room to maintain the desired thermal comfort level.

Mathematically, this feedback mechanism can be modeled as an optimization problem. Suppose a user adjusts the temperature to a new value $T_{preferred}$. The model takes this input and recalculates the optimal settings using an updated cost function that minimizes the error between the PMV prediction and the user-preferred comfort level. This is achieved through adaptive control, where the model adjusts based on changes in user-defined preferences:

$$J = \sum_{i=1}^{n} (PMV_i - PMV_{desired})^2 + \lambda (T_{AC} - T_{preferred})^2, \tag{3}$$

where J is the objective function that the model aims to minimize, λ is a weighting factor, T_{AC} is the current AC temperature setpoint, and $T_{preferred}$ is the temperature preferred by the user. The optimization balances maintaining a PMV value close to the desired range with meeting the user's preference.

Exhaustive Study of Comfort Temperatures

Mathematically, we conducted an exhaustive study across different AC setpoints and Air Velocities in the room to identify how comfort can be maintained under various conditions. The table provided (see Figure 3) shows the Predicted Mean Vote (PMV) values calculated for different combinations of AC setpoints and room air velocities.

AC Setpoint (°C)	Room Air Velocity (0.3 m/s)	0.4 m/s	0.5 m/s	0.6 m/s	(0.75 m/s	0.8 m/s	0.9 m/s
16	-2.8	-2.9	-3	-3	-3	-3	-3
17	-2.5	-2.6	-2.7	-2.8	-2.9	-3	-3.1
18	-2.2	-2.3	-2.4	-2.5	-2.6	-2.7	-2.8
19	-1.9	-2	-2.1	-2.2	-2.3	-2.4	-2.5
20	-1.6	-1.7	-1.8	-1.9	-2	-2.1	-2.2
21	-1.3	-1.4	-1.5	-1.6	-1.7	-1.8	-1.9
22	-1	-1.1	-1.2	-1.3	-1.4	-1.5	-1.6
23	-0.7	-0.8	-0.9	-1	-1.1	-1.2	-1.3
24	-0.4	-0.5	-0.6	-0.7	-0.8	-0.9	-1
25	-0.1	-0.2	-0.3	-0.4	-0.5	-0.6	-0.7
26	0.2	0.1	0	-0.1	-0.2	-0.3	-0.4
27	0.5	0.4	0.3	0.2	0.1	0	-0.1
28	0.8	0.7	0.6	0.5	0.4	0.3	0.2
29	1.1	1	0.9	0.8	0.7	0.6	0.5
30	1.4	1.3	1.2	1.1	1	0.9	0.8
31	1.7	1.6	1.5	1.4	1.3	1.2	1.1
32	2	1.9	1.8	1.7	1.6	1.5	1.4

Figure 3: Comfort PMV Values for Various AC Setpoints and Air Velocities. The green cells indicate a neutral comfort zone with PMV values between -0.5 and +0.5.

In the attached table, the numbers in the cells represent PMV values, where green cells (PMV between -0.5 and +0.5) indicate comfortable conditions. By studying these different settings, we have generated a detailed matrix of optimal AC temperatures and air velocities to maintain user comfort.

Machine Learning Model Adaptation

In the first version of the AC-Fan Sync AI model, the user-feedback is used to adjust the comfort temperature based on the changes he/she makes in the UX (app).

In the second version of the AC-Fan Sync AI model, we will include a learning mechanism so that the model adapts to user preferences over time by incorporating new data points obtained from the feedback mechanism. Each time a user provides feedback, the input (adjusted comfort temperature and air velocity) along with the observed PMV is added to the training dataset. This continuous stream of data will allow the model to refine its weights and parameters using techniques like online learning and reinforcement learning.

Future enhancements to the model

1. A reinforcement learning agent could be employed to maximize a reward function, where the reward is the user's satisfaction level.
2. The agent takes actions (adjusting air velocity or temperature), observes the environment (the user's feedback), and receives a reward based on how well the adjusted settings align with the user's preference.
3. Maximize cumulative rewards over time, thereby optimizing the model's ability to predict personalized comfort settings.

Reinforced Learning-based Model is under-development, but not part of Version 1

Mathematically, the update rule for reinforcement learning discussed above can be expressed as follows:

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)], \tag{4}$$

where $Q(s,a)$ represents the quality of taking action a in state s, α is the learning rate, r is the reward, and γ is the discount factor. This learning algorithm helps the model to better predict actions (AC setpoints and air velocities) that maximize user comfort.

Why we are using Neural Networks and not something simpler?

Despite having limited features in the first version, the neural network approach provides a mathematically robust solution to the thermal comfort prediction problem.

Here are some limitations with the alternatives:

- **Lookup Tables:** While simple to implement, they require $O(n^k)$ storage for k parameters with n discrete values each. Interpolation between points introduces additional error.
- **Linear Regression:** The model $y = X\beta + \epsilon$ cannot capture the inherent non-linearity of thermal comfort equations.
- **Rule-Based Systems:** They require explicit programming of all possible scenarios, leading to exponential growth in rule complexity.

The neural network architecture enables system evolution through Online Learning (i.e. in the future versions, the model can update its weights based on new data and we can deploy personalization therough Transfer Learning, where the initial weights trained on general comfort data can be fine-tuned for individual preferences. Also, we can handle non-linear relationships, adapt to user preferences (in version 2 of the algorithm), add additional variables (like skin temperature and heart rate in version 3 of the algorithm), and optimize multiple objectives simultaneously justifies its use despite its computational complexity. The system's mathematical framework ensures both accurate comfort prediction and efficient energy usage while maintaining the flexibility of the AC-Fan Sync AI algorithm to evolve over time, with user preferences and personalizations.